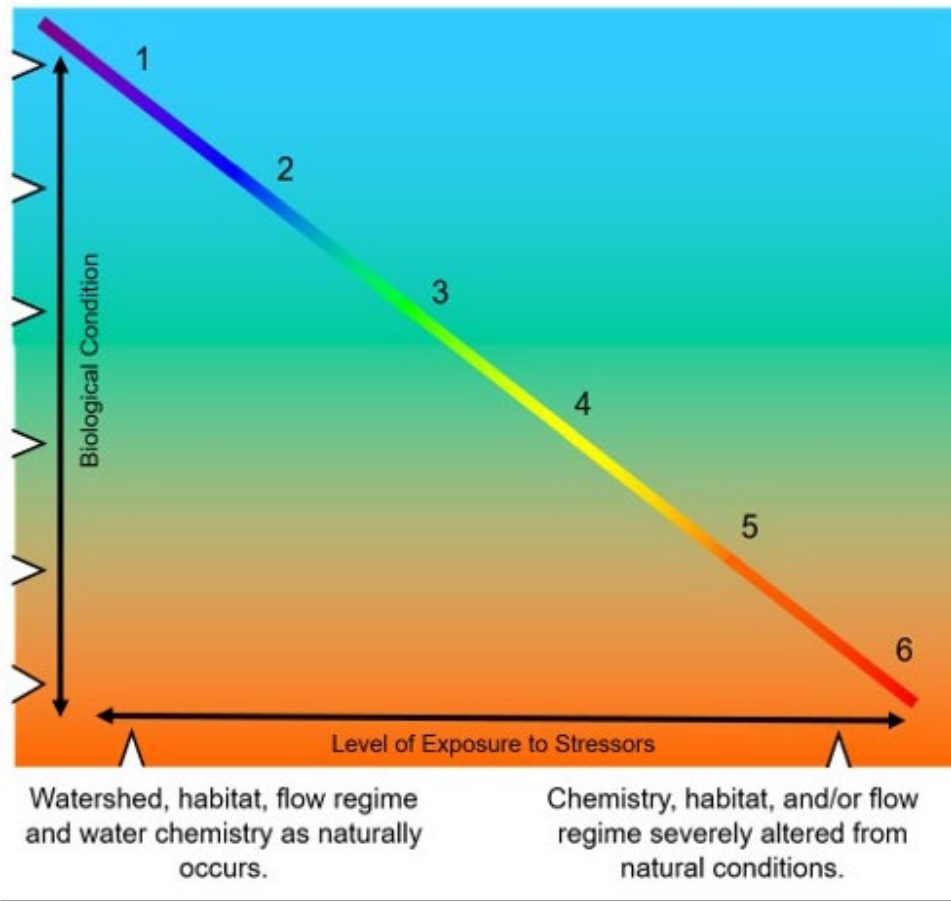




Quick tutorial: Lower Boise River Shiny app BCG calculator

tetrattech-wtr-wne.shinyapps.io/LowerBoiseBCGCalc/

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This R-based tool calculates Biological Condition Gradient (BCG) scores for macroinvertebrate and fish samples from the Lower Boise River mainstem.

Requirements

Users do not need R on their computers to run the Shiny app, nor do they need to have any familiarity with R.

To run the Shiny app you need:

- Internet access*
- Intermediate-level computer skills (need to be able to browse, upload and download files, unzip files)

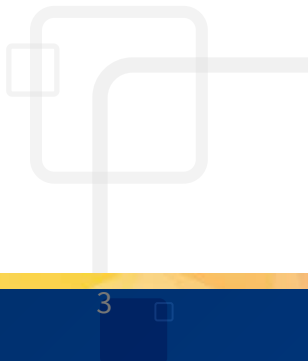
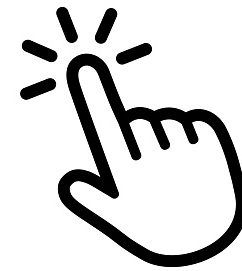
For those who prefer to work with R, the functions are carried out by the [BCGcalc](#) and [BioMonTools](#) R packages, which are available on GitHub and were developed by Erik W. Leppo from Tetra Tech (Erik.Leppo@tetrattech.com).

*If using the R code instead of the Shiny app, you won't need internet access if you have R installed on your computer and have downloaded all the necessary packages.

How the app works

Click buttons to

- Upload files
- Run the calculators
- Download results
- Unzip results



Workflow

1. Import File # 1 (.csv file with required fields)
2. Go to 'Prepare Input File' tab and run the File Builder for the selected community (bugs or fish)
3. Download the results; unzip the file; resolve unmatched taxa if necessary
4. Import File # 2 (BCG_TaxaTranslator_Results_wAttributes.csv output from the File Builder)
5. Run BCG calculation for the selected community (bugs or fish)
6. Download the results; unzip the file; interpret results

Time saver tip

Some internet browsers, like Google Chrome and Mozilla Firefox, allow you to change settings so that you can select which folder to download files to (versus files going to the default Download folder).

- Mozilla Firefox: go to Settings > General > Files and Applications > Check box for “always ask where to save files.”
- [Google Chrome file saving tip](#) [PDF]

Step 1: Set up a directory

This is the folder where you'll be saving your files



Step 2: Prepare Input File

The File Builder joins the taxon name in the input file with the 'original taxalD' column in the [Taxa Translator table](#) and performs two functions:

- Translates the original taxon name to the proper Operational Taxonomic Unit (OTU) for the calculation (for example, collapses worms to Oligochaeta).
- Adds in new columns with taxa attributes and hierarchy (e.g., Order, Family) that are needed for the metric calculations.

[BioMonTools_SupportFiles/data/taxa_official/LowerBoise at main · leppott/BioMonTools_SupportFiles · GitHub](#)

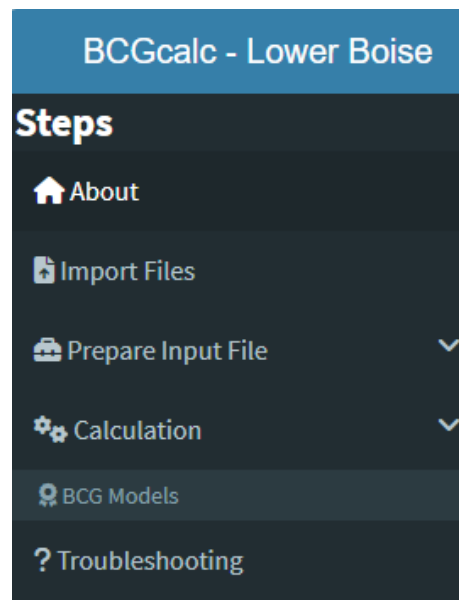
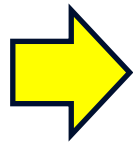
File Builder input file – required content

Required fields for all calculations. Column names are flexible (use drop-down menus to match).

Column Name	Fish	Bugs	Data Type	Allowable entries	Description
locationId	x	x	text or numeric	Flexible	Site identifier
activityYear	x	x	numeric	Positive integer	Year of the sampling event
activityDate		x	date	Date, preferably in this format: MM/DD/YYYY	Example: 10/26/1996. Not required for fish because the two passes are sometimes done on different dates.
SampleID	x	x	text	Flexible	Unique sample identifier
TaxaID	x	x	text	Flexible	Taxon name
Count	x	x	numeric	Non-negative real numbers; should not be blank	Number of individuals
Index_Class	x	x	text	See list below (must match exactly)	Tells R which rules and metrics to apply, based on class
Length	x		numeric	Non-negative real numbers; blanks are OK	Length, units = mm
deltCount	x		numeric	Non-negative real numbers; blanks are OK	number of individuals with Deformity, Erosion, Lesion or Tumor (DELT) anomalies
density_m2		x	numeric	Non-negative real numbers; decimals and blanks are OK	number of individuals per meter squared (count x subsample correction factor x area correction factor)

Step 3: Import File # 1

Must be a comma-separated file (CSV). It can include **data for multiple samples, sites and Index Classes** (versus being limited to one sample at a time).
File uploads are limited to a maximum of 200 MB in size.



Load File
Only comma-separated or tab-separated files.

Select file parameters

Separator
☒ Comma
☐ Tab

Choose file to upload
Browse... BugBCG_FileBuilder_Test1_20240725.csv
Upload complete

The 'separator' allows the user to upload different file formats (e.g., csv, tsv, or txt).
Files for all operations will be uploaded through this interface.
File uploads are limited to a maximum of 200 MB in size.

A table is shown below after data is loaded.

Show 5 entries

Table. Imported data.

	SampleID	Taxonomic.Name	Count	Index_Class	Index_Name
	All	All	All	All	All
1	Test1	Acerpenna pygmaea	4	bug4	MN_BCG
2	Test1	Atherix variegata	1	bug4	MN_BCG
3	Test1	Brachycentrus numerosus	4	bug4	MN_BCG
4	Test1	Caenis	5	bug4	MN_BCG
5	Test1	Calopteryx aequabilis	3	bug4	MN_BCG

Showing 1 to 5 of 257 entries

Previous 1 2

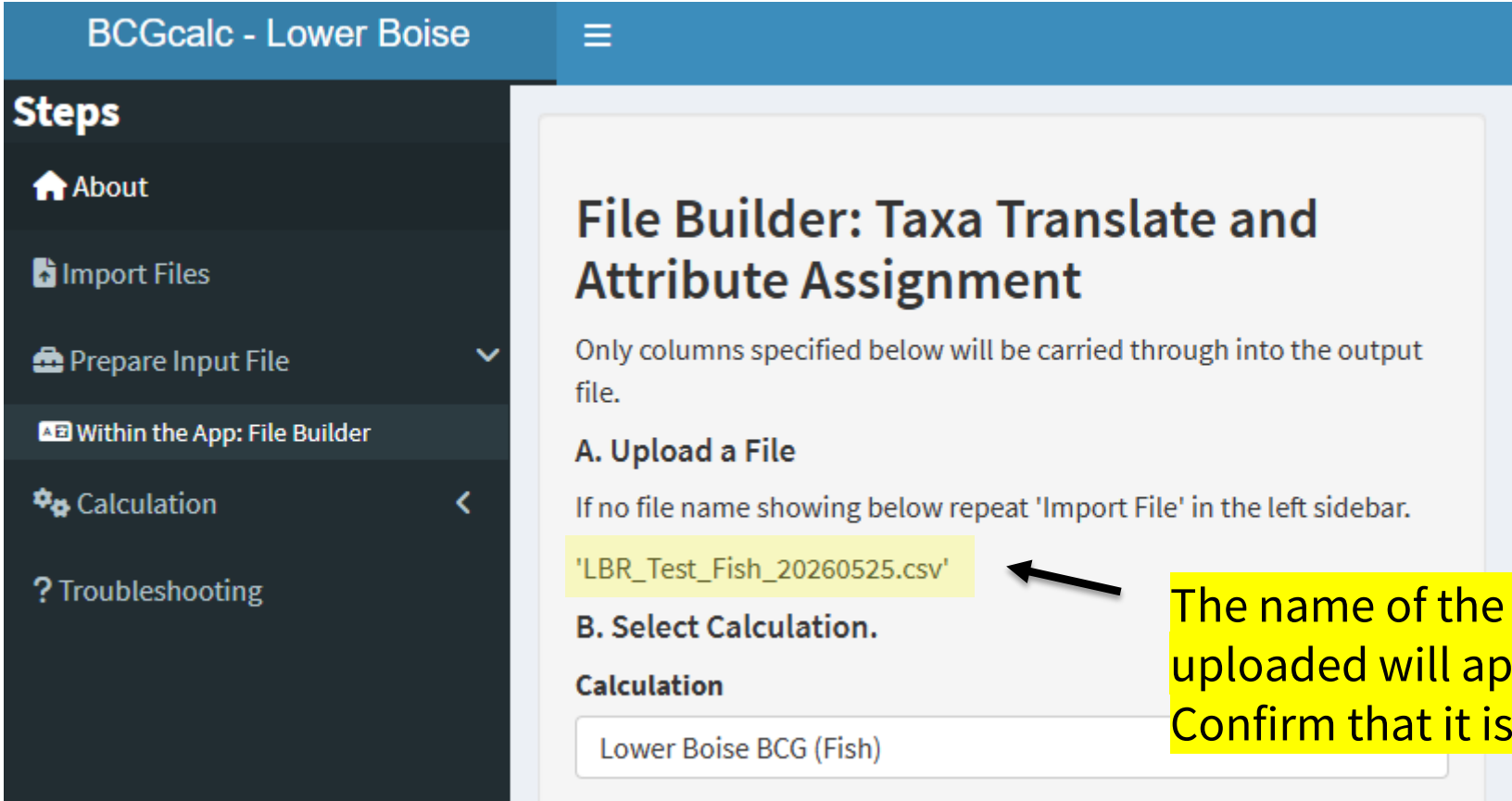
1. Browse to the file

2. Status bar will tell you when the upload is complete

3. A file preview will appear. Check to make sure everything looks ok.

Step 4: Run File Builder

IMPORTANT! The BCG models are calibrated for specific collection and processing protocols. Make sure you are running the BCG calculator on appropriate data.



BCGcalc - Lower Boise

Steps

- About
- Import Files
- Prepare Input File
- Within the App: File Builder**
- Calculation
- Troubleshooting

File Builder: Taxa Translate and Attribute Assignment

Only columns specified below will be carried through into the output file.

A. Upload a File

If no file name showing below repeat 'Import File' in the left sidebar.

'LBR_Test_Fish_20260525.csv'

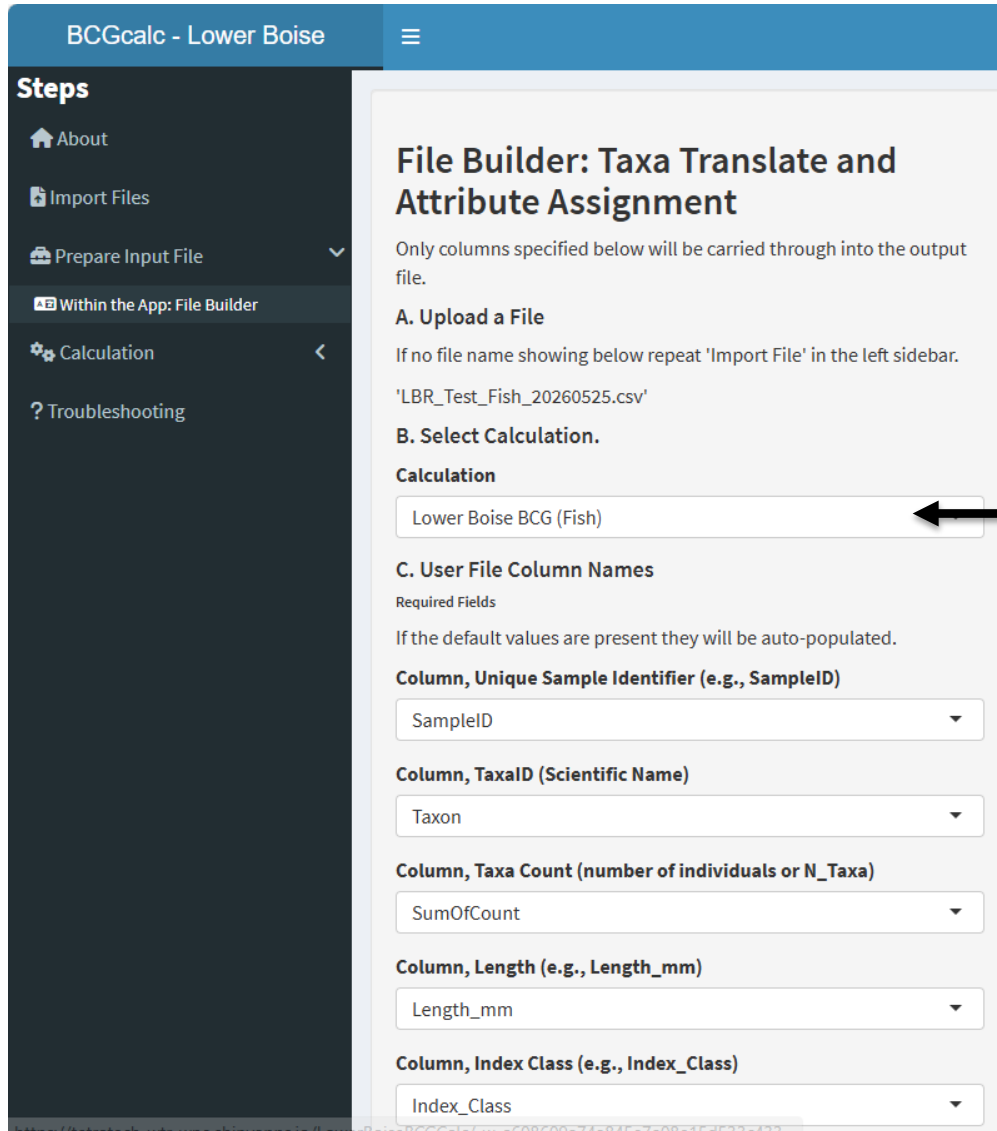
B. Select Calculation.

Calculation

Lower Boise BCG (Fish)

The name of the file that you uploaded will appear here. Confirm that it is the correct file.

File Builder – enter required fields



Side note:

- **INDEX_NAME** (*LBR_Bugs_BCG_v1* or *LBR_Fish_BCG_v1*) is also a required field but is autopopulated by R when you run the File Builder.

Select calculation

Enter the **required fields**

Drop-down boxes allow for flexibility with column names (match with the appropriate columns in the input file).

File Builder – optional fields

Columns to Keep in Output

N_ANOMALIES |

Index_Class

SampleID

Taxon

SumOfCount

Length_mm

All input and output files will be available in a single zip file.

 Download Results

OPTIONAL fields can be included as well and will be carried through into the File Builder output file.

File Builder – Run Operation

Index_Class ▾

Select Additional Required Fields (which vary by Calculation – see table on right).

Columns need to be named exactly as shown on the right. Optional fields (e.g., SiteID, collection date) can be selected as well. IMPORTANT! Do not include the same field twice and do not include Life Stage or other fields that might cause a taxon to occur in more than one row for a given sample (which could lead to double-counting of that taxon in the richness metrics).

Columns to Keep in Output

GP.RR

D. Run Operation

This button will merge the user file with the official taxa file

Run Operation

E. Output

All output files will be available in a single zip file.

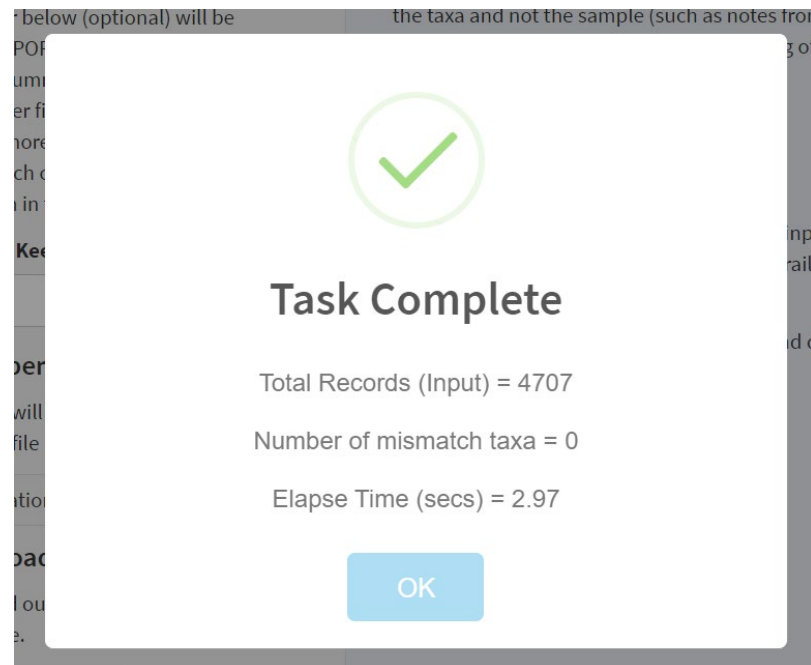
Download Results

Click on the Run Operation button

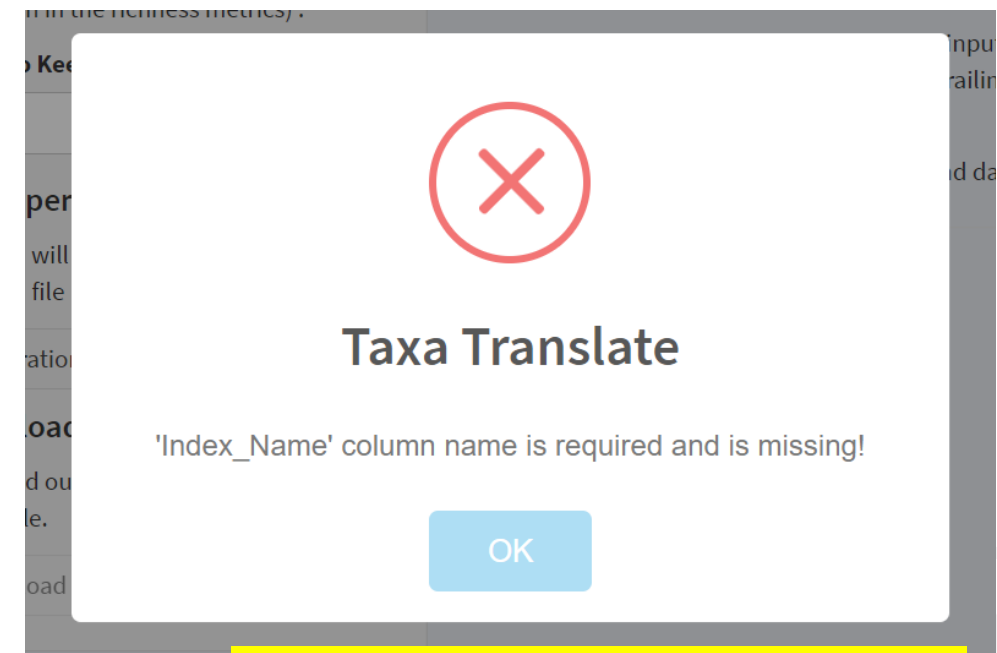
File Builder – prompt box

When the File Builder has finished running, **a prompt box will appear**. It will notify you if your File Builder run was successful or failed.

Success!



Failure!

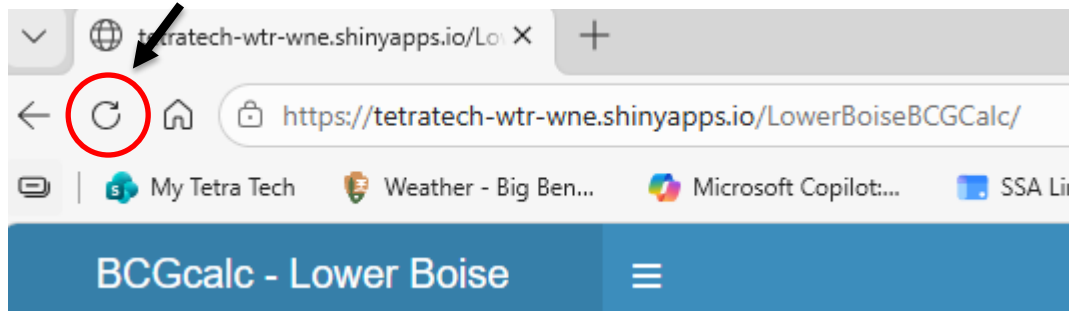


Will alert you to missing columns

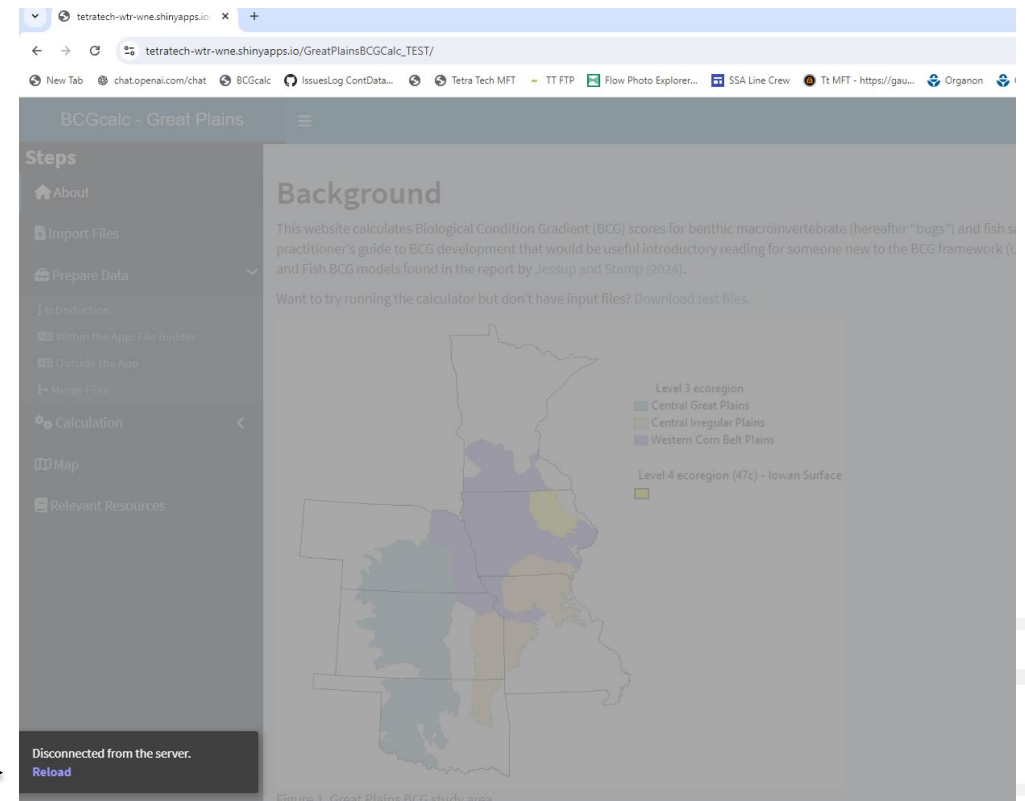
Refresh the Shiny app

In some cases, the screen will gray out (meaning the Shiny app has crashed or timed out) and you'll have to refresh or reload the app, investigate potential problems with your input file, and try again.

Refresh button (upper left corner)



Reload if app crashes
(gray screen)



File Builder – Download Results

Index_Class ▾

Select Additional Required Fields (which vary by Calculation – see table on right).

Columns need to be named exactly as shown on the right. Optional fields (e.g., SiteID, collection date) can be selected as well. IMPORTANT! Do not include the same field twice and do not include Life Stage or other fields that might cause a taxon to occur in more than one row for a given sample (which could lead to double-counting of that taxon in the richness metrics).

Columns to Keep in Output

GP.RR

D. Run Operation

This button will merge the user file with the official taxa file

Run Operation

E. Download Output

All input and output files will be available in a single zip file.

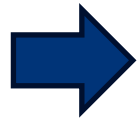
Download Results



The 'Download Results' button will become clickable when the files are ready to download. Click the button and save the zipped file to the directory. The file will either automatically go to the Download folder or, if you use the Time Saver tip with Google Chrome or Mozilla Firefox (see slide 6), you can browse to the desired folder and save it there.

Unzip File Builder results

Unzip
the file



A 'results'
folder will
appear



Subfolders within the 'results' folder



_user_input_Bugs



_user_input_Fish



reference



results_Bugs_BCG



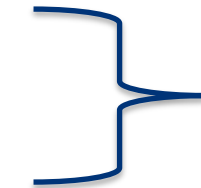
results_Fish_BCG



Original input files



Taxa translator, taxa attribute and fish age
class files



File Builder & Calculation results

Users must unzip/extract the files or else the files won't be seen as an option to upload for the next step. After files are unzipped, the original zipped files can be deleted if desired.

Unzipping files

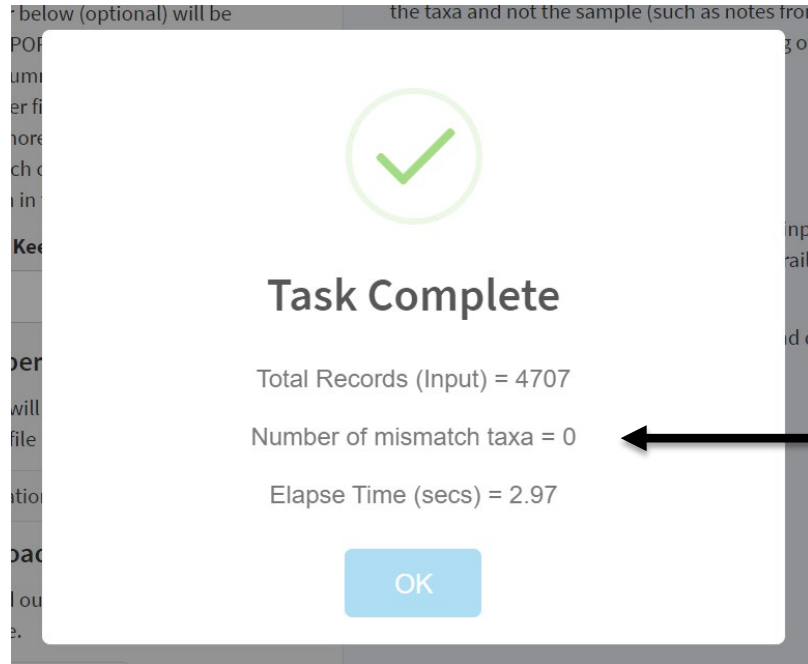
For whatever reason, 7zip (free) seems to work better than the Microsoft default zip extractor with our Shiny apps (but if you can't get 7zip, the Microsoft option should work ok).

7zip (<https://www.7-zip.org/>)

File Builder outputs

Output file	Description
BCG_TaxaTranslator_modifications	shows list of taxa in the input file; to see which taxa names were modified during the Taxa Translation process for reasons other than capitalization scheme, filter the Modified_woCAPS column = TRUE
BCG_TaxaTranslator_nonmatches	taxa in your input file that don't match with the taxa list in the Taxa Translator table
BCG_TaxaTranslator_Results_wAttributes	input file is now ready to go! The taxa names have been translated to the BCG OTU and attributes and hierarchy added in. Import this file into the Shiny app and run the BCG calculator
BCG_TaxaTranslator_sources	shows which calculation was selected, which Operational Taxonomic Unit (OTU) column was used, and which versions of the Taxa Translator and Attribute tables were used

Non-matching taxa



What if there are non-matching taxa?

Open the 'BCG_TaxaTranslator_nonmatch' file and view the list of non-matching taxa. If the non-matches are due to a misspelling, extra space or other small issue, fix the non-matching taxa in your input file, re-upload the file via the 'Import File' tab, and rerun the File Builder (and to prevent confusion, delete the first set of output files with the non-matching taxa). If the non-matching taxa are legitimate taxa that need to be added to the Taxa Translator table, notify Jen.Stamp@tetratech.com and Tetra Tech will update the Taxa Translator table as soon as resources permit.

Example



results_Fish_BCG



 BCG_TaxaTranslator_modifications.csv

 BCG_TaxaTranslator_nonmatches.csv

 BCG_TaxaTranslator_Results_wAttributes.csv

 BCG_TaxaTranslator_sources.csv

Step 5: Import File #2

Import the newly created 'BCG_TaxaTranslator_Results_wAttributes' file

Example



results_Fish_BCG



BCG_TaxaTranslator_modifications.csv

BCG_TaxaTranslator_nonmatches.csv

BCG_TaxaTranslator_Results_wAttributes.csv

BCG_TaxaTranslator_sources.csv

BCGcalc - Lower Boise



Steps

About

Import Files

Prepare Input File

Calculation

Troubleshooting

Load File

Only comma-separated or tab-separated files.

Select file parameters

Separator

☒ Comma

☐ Tab

Choose file to upload

Browse...

BCG_TaxaTranslator_Results_wAttributes.csv

Upload complete

The 'separator' allows the user to upload different file formats (e.g., csv,

2. Status bar will tell you when the upload is complete

A table is shown below after data is loaded.

Show 5 entries

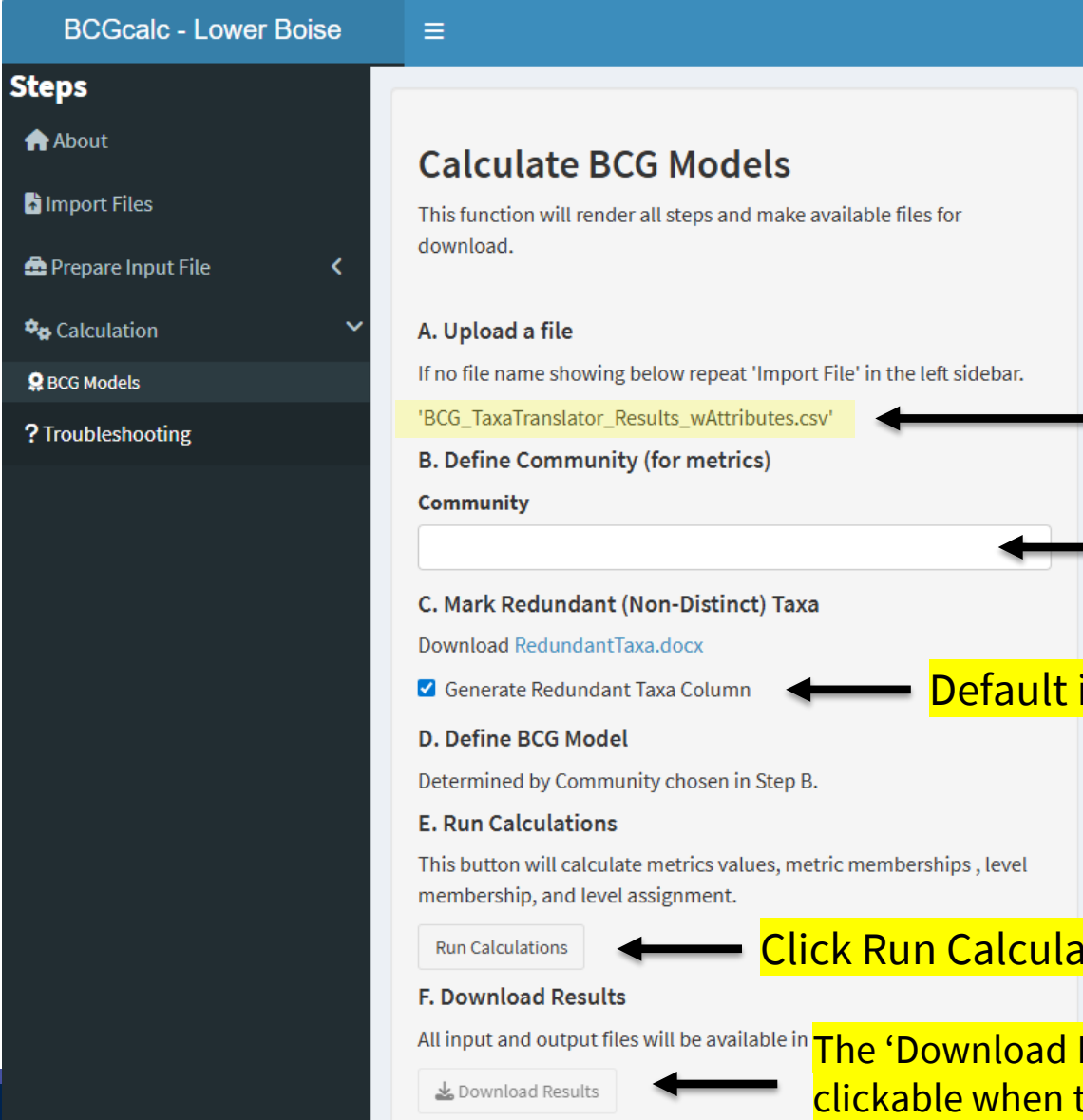
Search:

Table. Imported data.

	SampleID	TaxaID	N_Taxa	file_taxatrans	file_attributes
	All	All	A	All	All
1	Test1	Acari	4	LowerBoise_Bugs_Taxa_Translator_20260508.csv	LowerBoise_Bugs_Attributes_202
2	Test1	Acari	8	LowerBoise_Bugs_Taxa_Translator_20260508.csv	LowerBoise_Bugs_Attributes_202
3	Test1	Acari	5	LowerBoise_Bugs_Taxa_Translator_20260508.csv	LowerBoise_Bugs_Attributes_202
4	Test1	Acentrella insignificans	2	LowerBoise_Bugs_Taxa_Translator_20260508.csv	LowerBoise_Bugs_Attributes_202

3. A file preview will appear. Check to make sure everything looks ok.

Step 6: Run desired calculation



The screenshot shows the BCGcalc - Lower Boise web application. A large yellow arrow points to the 'Calculation' step in the left sidebar. The main content area displays the 'Calculate BCG Models' section with the following steps:

- A. Upload a file**
This function will render all steps and make available files for download.
If no file name showing below repeat 'Import File' in the left sidebar.
The file name 'BCG_TaxaTranslator_Results_wAttributes.csv' is displayed in a yellow box. An annotation points to it: "The name of the file that you uploaded will appear here. Confirm that it is the correct file."
- B. Define Community (for metrics)**
Community
A text input field is shown. An annotation points to it: "Select model"
- C. Mark Redundant (Non-Distinct) Taxa**
Download [RedundantTaxa.docx](#)
☒ Generate Redundant Taxa Column. An annotation points to it: "Default is for R/Shiny app to mark redundant taxa"
- D. Define BCG Model**
Determined by Community chosen in Step B.
- E. Run Calculations**
This button will calculate metrics values, metric memberships, level membership, and level assignment.
A 'Run Calculations' button is shown. An annotation points to it: "Click Run Calculation button"
- F. Download Results**
All input and output files will be available in
A 'Download Results' button is shown. An annotation points to it: "The 'Download Results' button will become clickable when the files are ready to download"

BCG calculator workflow

- ✓ Mark non-distinct taxa (excluded from richness metrics)
- ✓ Calculate metric values each sample
- ✓ Calculate metric membership values for each sample
- ✓ Assign membership in each BCG level

Unzip Calculation results

Unzip
the file



Output files will
be added to the
existing folders



results_Bugs_BCG



results_Fish_BCG



 _BCG_RESULTS.csv
 _BCG_RESULTS.html
 BCG_1markexcl.csv
 BCG_2metval_BCG.csv
 BCG_2metvall_all.csv
 BCG_3metmemb.csv
 BCG_3metmemb_xtab_METRICSORT.csv
 BCG_3metrules.csv
 BCG_4levmemb.csv
 BCG_5levassign.csv
 BCG_6levflags.csv
 BCG_6metflags.csv
 BCG_TaxaTranslator_modifications.csv
 BCG_TaxaTranslator_nonmatches.csv
 BCG_TaxaTranslator_Results_wAttributes.csv
 BCG_TaxaTranslator_sources.csv

Calculation
output files

File
Builder
files

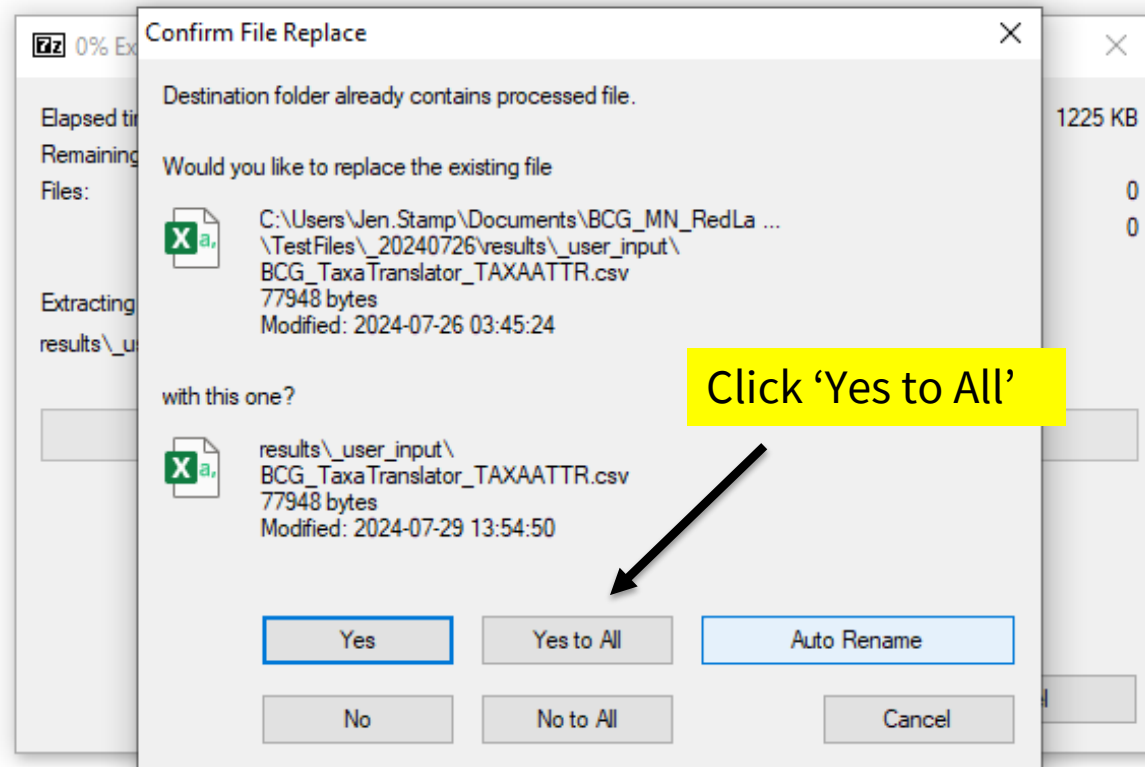
**After files are unzipped, the original zipped files can be
deleted if desired.**

Fish outputs will have two additional outputs

- Fish_AgeClass_xtab
- Fish_AgeClass_pClasses

Unzip Calculation results

You may receive a prompt like this when you unzip the Calculation results file.



Questions? Comments?

We're always looking for feedback on ways we can improve the Shiny app.

Jen.Stamp@tetratech.com

